Concept

$$132 \times \frac{20}{100} + 132$$

$$132\left(1 + \frac{20}{100}\right)$$

$$132\left(\frac{120}{100}\right)$$

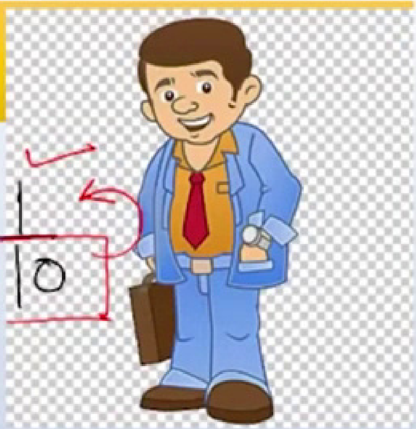
Increase a certain number by 20%?

Increase a c[illegible]umber by 25%?

Increase a certain number by 33.33%?

Note :

Concept



$10\%\uparrow \quad \times \frac{110}{100}$

$10\%\downarrow \quad \times \frac{90}{100}$

$10\%\uparrow = \frac{10}{100} = \frac{1}{10}$

$\times \frac{11}{10}$

Increase a certain number by 20%?

$\times \frac{120}{100}$

Increase a certain number by 33.33%?

Increase a certain number by 25%?

$\times \frac{125}{100} \Rightarrow 25\%$

Note :

$\frac{1}{2} = 50\%$	$\frac{1}{8} = 12.50\%$ or $12\frac{1}{2}\%$
$\frac{1}{3} = 33.33\%$ or $33\frac{1}{3}\%$	$\frac{1}{9} = 11.11\%$ or $11\frac{1}{9}\%$
$\frac{1}{4} = 25\%$	$\frac{1}{11} = 9.09\%$ or $9\frac{1}{11}\%$
$\frac{1}{5} = 20\%$	$\frac{1}{12} = 8.33\%$ or $8\frac{1}{3}\%$
$\frac{1}{6} = 16\frac{2}{3}\%$	$\frac{1}{14} = 7.14\%$ or $7\frac{1}{7}\%$
$\frac{1}{7} = 14.28\%$ or $14\frac{2}{7}\%$	$\frac{1}{15} = 6.66\%$ or $6\frac{2}{3}\%$

NOTE :

$$\frac{1}{3} = 33\frac{1}{3}\% \Rightarrow \frac{2}{3} = 66\frac{2}{3}\%$$

$$\frac{1}{8} = 12.5\% \Rightarrow \frac{3}{8} = 37.5\%$$

Type 1 :

TIME'S UP!

1

Q.1 A man buys a cycle for Rs.1400 and sells it at a loss of 15%. What is the selling price of the cycle?

2

Q.2 On selling an article for Rs.960, there is a loss of 4%. The cost price of that article is ?

$$\boxed{S.P = CP \times M.F}$$

$$SP = 1400 \times \frac{85}{100}$$

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1 An article is sold at Rs.640 at a profit of 14.28%,Find the Cost Price ?

Q.3

TITA

$+\frac{1}{7}$

$SP = CP \times M.F$

$\cancel{640}^{80} = CP \times \frac{\cancel{8}}{7}$

560

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CLICK TO RESE

TIME'S U

4 Type 2 :

A person buys a radio for Rs.1030 and he spent Rs. 50 on its repairs. If he sold it for rs1200, find the profit percent.

CP SP

$5 \longrightarrow 6$

$\frac{+1}{5} \times 100 = 20\%$

$\boxed{11.11\% = \frac{+1}{9}}$

$CP = 1030$
$+50$
$CP = \boxed{1080} \longrightarrow SP = 1200$

$\frac{CP}{SP} = \frac{\cancel{1080}}{\cancel{1200}} = \frac{9}{10}$

CLICK TO RESET

TIME'S U

Note :

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If the cost price of 15 tables be equal to the selling price of 20 tables, the loss per cent is 25%

5

$$CP \times 15 = SP \times 20$$

$$\frac{CP}{SP} = \frac{\cancel{20}}{\cancel{15}} = \frac{4}{3}$$

If the cost price of 50 to the selling price o the profit per cent is :

CLICK TO RESET

TIME'S UP!

6

if the cost price of 50 ranges i sequal to the selling price of 40 oranges , then the profit percent is:

$$-\frac{1}{4} = 25\%$$

If the cost price of 15 tables be equal to the selling price of 20 tables, the loss per cent is

____

7

CLICK TO RESET

TIME'S UP!

If the cost price of 50 oranges is equal to the selling price of 40 oranges, then the profit per cent is :

____

8

$$CP \times 50 = 40 \times SP$$

$$\frac{CP}{SP} = \frac{40}{50} = \frac{4}{5}$$

$$\frac{1}{4} = 25\%$$

Type 3 : 9

A bought toffees at a rate of 15per rupee and sold them at the rate of 12 for a rupee. His gain percent?

ASLI

$CP_1 = \frac{1}{\cancel{15}} \times \cancel{60}$ → 4

$SP = \frac{1}{\cancel{12}} \times \cancel{60}$ → 5

$= 25\%$

$$\frac{CP = \frac{1}{15}}{SP = \frac{1}{12}} = \frac{1}{\cancel{15}} \times \frac{\cancel{12}}{1} = \frac{4}{5}$$

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CLICK TO RESET

TIME'S UP!

A person buys 8 oranges for Rs.15 and sells them at 10 for Rs.18. What does he gain or lose percent?

10

$$CP_1 = \frac{15}{8} \times [40]$$

$$SP_1 = \frac{18}{10} \times [40]$$

$$\frac{CP}{SP} = \frac{\cancel{15}^{5}}{\cancel{8}_{4}} \times \frac{\cancel{10}^{5}}{\cancel{18}_{6}} = \frac{25}{24}$$

$$25 \longrightarrow 24$$

$$\frac{-1}{25} \times 100 = -4\%$$

A man purchases a certain number of apples at 3 per rupee and the same number of apples at 4 per rupee. He mixes them together and sells them at 3 per rupee. What is his gain or loss per cent.?

$$CP_1 = \frac{1}{\cancel{3}} \times \cancel{12}^{4} + \frac{1}{\cancel{4}} \times \cancel{12}^{3} = 7$$

$$SP = \frac{1}{\cancel{3}} \times \cancel{24} = 8$$

$$\frac{+1}{7} = 14.28\%$$

$$\frac{CP}{SP} = \frac{\frac{1}{3} + \frac{1}{4}}{\frac{1}{3} \times 2}$$

CLICK TO RESET

TIME'S UP!

Type 4 :

By selling an article for Rs. 240, a man incurs a loss of 10%. At what price should he sell it to make a profit of 20%?

12

$SP = CP \times M.F$

$240 = CP \times \frac{90}{100}$

$CP = \left[\frac{240 \times 100}{90}\right]$

$\Rightarrow \frac{240}{90} \times 120$

$SP = \left[\frac{240 \times 100}{90}\right] \times 120$

CLICK TO RESET

TIME'S U

By selling an article for Rs. 480 a person looses 20%. For what amount should he sell it to make a profit of 20% ?

12A

$$\Rightarrow \frac{480}{80} \times 120$$

CLICK TO RESET

TIME'S UP!

Type 4 :

A man sold two bicycles at Rs.99 each. He sold one at a loss of 10% and other at a profit of 10%. Find his profit or loss percentage?

13

00:13

II PAUSE

Note :

Type 4 :

A man sold two bicycles at Rs.99 each. He sold one at a loss of 10% and other at a profit of 10%. Find his profit or loss percentage?

14

$SP_1 = 99$

$\cancel{99}^{11} = CP \times \frac{\cancel{90}}{100}$

$\boxed{CP = 110}$

$SP_2 = 99$

$\cancel{99}^{9} = CP \times \frac{\cancel{110}}{100}$

$\boxed{CP = 90}$

$\boxed{S.P = CP \times M.F}$

$SP = 198$

$CP = 200$

$\frac{-2}{200} \times 100$

-1%

CLICK TO RESET

TIME'S UP!

Note :

Type 4 :

A man sold two bicycles at Rs.99 each. He sold one at a loss of 10% and other at a profit of 10%. Find his profit or loss percentage?

15

$\boxed{S.P = CP \times M.F}$

$SP_1 = 99$

$99 = CP \times \frac{90}{100}$

$CP = 1$

$SP_2 = 99$

$99 = CP \times \frac{110}{100}$

$\boxed{CP = 90}$

$SP = 198$

$CP = 200$

$\frac{-2}{200} \times 100$

-1%

$\boxed{-\frac{x^2}{100} =}$

Ans.

Note : ① $SP_1 = SP_2$ ② $P\% = L\% = x\%$

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A man sold two tyres at Rs.2400 each. He sold one at a loss of 20% and other at a profit of 20%. Find his profit or loss percentage?

16

$$\left.\begin{array}{l} SP_1 = SP_2 \\ P\% = L\% \end{array}\right\}$$

$$-\frac{x^2}{100}$$

$$-\frac{400}{100} = -4\%$$

CLICK TO RESET

TIME'S UP!

Type 5 :

The profit earned by selling an article for Rs832 is equal to the loss incurred when the same article is sold for Rs 448. What should be the sale price of the article for making 50 percent profit?

17

$832 - CP = CP - 448$

$CP = 640 \times 1.5$

$$\frac{832 + 448}{1280} / 2$$

$640 \times 1.5 = 960$

Note :

CLICK TO RESET

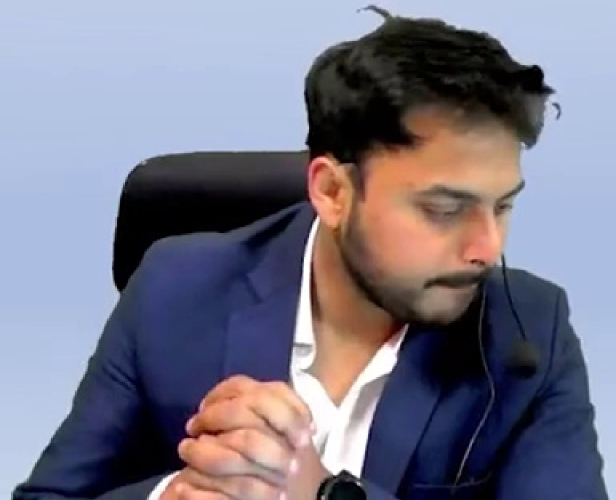
TIME'S UP!

The profit earned by selling an article for Rs482 is equal to the loss incurred when the same article is sold for Rs 318.What should be the selling price of the article for making 30 percent profit?

18

00:11

II PAUSE



The profit earned by selling an article for Rs482 is equal to the loss incurred when the same article is sold for Rs 318. What should be the selling price of the article for making 30 percent profit?

CLICK TO RESET

TIME'S UP!

19

400×1.3

520

TIME'S UP!

Type 6 :

I sold a book at a profit of 12 percent had I sold it for Rs.18 more 18percent would have been gained find the CP

20

$$SP = CP \times M.F$$

$$SP = 100 \times \frac{112}{100}$$

$$\Rightarrow (SP + 18) = 100 \times \frac{118}{100}$$

$$-1$$

$$-11$$

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Note :

TIME'S UP!

Type 6 :

I sold a book at a profit of 12 percent had I sold it for Rs.18 more 18percent would have been gained find the CP

21

CP	SP
100	112
	118

6 = 18

$\boxed{1 = 3}$

300 Ans

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Note :

A man sold a horse at a loss of 7 %. Had he been able to sell it at gain of 9 %, it would have fetched Rs. 64 more than it did. What was the cost price?

22

CP → SP

100 → 93

100 → 109

400

$16 = 64$

$\boxed{1 = 4}$

CLICK TO RESET

TIME'S UP!

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Question Of the Day

A sold his car B at a profit of 20% and B sold it to C at a profit of 10%. C sold it to mechanic at a loss of 9.09%.Mechanic spent 10% of his purchasing price and ~~then~~ sold it at a profit of 8.33% to A once again. what is the loss of Anna?

23% B.43% C.50% D.40%

23

$$A \xrightarrow{+\frac{1}{5}} B \xrightarrow{+\frac{1}{10}} C \xrightarrow{-\frac{1}{11}} \text{Mech}$$

$$1 \times \frac{6}{5} \times \frac{11}{10} \times \frac{10}{11} = \boxed{120} + 12 = 132 \times \frac{13}{12} = 143$$

23%

~~43~~ %

CLICK TO RESET

TIME'S UP!

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Type 8 :

Rahul sells a bicycle to Banu at a profit of 25%. Banu sells it to Sona at a profit of 20%. If sona pays Rs.3000 for it, then the cost price of the bicycle for Rahul is

24

Asli Method :

$R \xrightarrow{25\%} B \xrightarrow{+20\%} S = 3000$

$\frac{1}{4}$ $\frac{1}{5}$

$$x \times \frac{125}{100} \times \frac{120}{100} = 3000$$

$$x \times \frac{5}{4} \times \frac{6}{5} = \cancel{3000}\ 100$$

2000 Ans

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CLICK TO RESET

TIME'S UP!

Sum of CP's of two cows is Rs. 39, 000. Both the cows are sold at a profit of 20% and 40% respectively with their SP's being the same. What is the difference of CP's of both the cows? Type 7 26

1) Rs. 3,000 2) Rs. 2, 000 3) Rs. 1, 500 4) Rs. 2, 500 5) None of these

$CP_1 + CP_2 = 39000$

$13 \rightarrow 39000$

$\boxed{1 \rightarrow 3000}$

$\boxed{SP_1 = SP_2}$

$CP_1 \times \cancel{12}^{6} = CP_2 \times \cancel{14}^{7}$

$\boxed{\frac{CP_1}{CP_2} = \frac{7}{6}}$

CLICK TO RESET

'S UP!

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25

Type 7 :

A person bought two watches for Rs.480. He sold one at a loss of 15% and the other at a gain of 19% and he found that each watch was sold at the same price. Find the cost price of the two watches.

$CP_1 + CP_2 = 480$

$CP_1 = x$

$CP_2 = (\ldots - x)$

SP_2

$12 \rightarrow 480$

$\boxed{1 \rightarrow 40}$

Asli Method :

$SP_1 = SP_2$

$CP_1 \times \cancel{85}^{5} = CP_2 \times \cancel{119}^{7}$

$\boxed{\frac{CP_1}{CP_2} = \frac{7}{5}}$

$7 \Rightarrow 280$

$5 \Rightarrow 200$

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